Practical–2: Study of Different Types of Network Cables

Aim

To study different types of network cables and understand their construction, characteristics, advantages, disadvantages, and applications. Also, to construct an Ethernet straight-through and crossover cable.

1. Types of Network Cables

The major types of network cables studied are:

1. Unshielded Twisted Pair (UTP) Cable
2. Shielded Twisted Pair (STP) Cable
3. Coaxial Cable
4. Fibre Optic Cable

Types of Network Cables – Details

Cable Type	Category/Type	Typical Data Rate	Advantages	Disadvantages	Applications
UTP	Cat 3	Up to 10 Mbps	Low cost, easy to install, flexible	More susceptible to EMI and noise	10BASE-T Ethernet
UTP	Cat 5	Up to 100 Mbps	Low cost, easy installation	Older standard, more interference	Fast Ethernet
UTP	Cat 5e	Up to 1 Gbps	Affordable, widely available	Less protection from EMI	Fast Ethernet, Gigabit Ethernet
STP	Cat 6 / Cat 6a	Up to 10 Gbps	Shielding reduces noise and interference	More expensive, difficult to install	Gigabit Ethernet, 10G Ethernet
SSTP	Cat 7	Up to 10 Gbps	Excellent shielding, high performance	Expensive and difficult to install	Gigabit Ethernet, 10G Ethernet

Cable Type	Category/Type	Typical Data Rate	Advantages	Disadvantages	Applications
Coaxial	RG-59, RG-6, RG-11	Depends on application	Good shielding, reliable	Bulky, less flexible	TV networks, broadband Internet
Fibre Optic	Single-mode	Very high speed	High bandwidth, long distance, secure, immune to EMI	Expensive, requires skilled installation	WANs, data centres, backbone networks
Fibre Optic	Multimode	Very high speed	High bandwidth, suitable for shorter distances	More distance-limited than single-mode	LANs, data centres

The cable characteristics and applications above are based on the cable table in the provided experiment document.

2. UTP Cable

UTP (Unshielded Twisted Pair) consists of pairs of insulated copper wires twisted together. The twisting helps reduce electromagnetic interference and crosstalk.

Advantages

- Low cost
- Easy to install
- Lightweight and flexible
- Small overall diameter

Disadvantages

- More susceptible to electromagnetic interference than shielded cables
- Limited transmission distance for high-speed Ethernet

Applications

- Ethernet LANs
- Telephone networks
- Fast Ethernet

- Gigabit Ethernet

3. STP Cable

STP (Shielded Twisted Pair) is similar to UTP but contains additional metallic shielding around the wire pairs or cable.

Advantages

- Better protection against EMI
- Reduced interference and crosstalk
- Suitable for electrically noisy environments

Disadvantages

- More expensive than UTP
- More difficult to install
- Requires proper grounding

Applications

- Gigabit Ethernet
- 10-Gigabit Ethernet
- Industrial networks
- Data centres

4. Coaxial Cable

A coaxial cable contains a central conductor, insulating layer, metallic shield, and outer protective jacket.

Common Types

- RG-59
- RG-6
- RG-11

Advantages

- Good resistance to interference
- Reliable signal transmission
- Suitable for broadband applications

Disadvantages

- Bulky compared with twisted-pair cable

- Less flexible
- Installation can be more difficult

Applications

- Television networks
- Cable Internet
- CCTV systems
- RF communication

5. Fibre Optic Cable

A fibre optic cable transmits information using pulses of light through glass or plastic optical fibre rather than electrical signals.

Types

1. Single-mode fibre
2. Multimode fibre

Advantages

- Very high bandwidth
- High transmission speed
- Long-distance communication
- Immune to electromagnetic interference
- High security

Disadvantages

- Higher installation cost
- Requires specialised tools and skilled technicians
- Fibre can be more delicate than copper cable

Applications

- Internet backbone
- Data centres
- Long-distance communication
- Telecommunications
- High-speed networks

b) Making an Ethernet Straight-Through / Crossover Cable

Tools and Materials Required

- CAT5/CAT5e Ethernet cable
- RJ45 connectors – 2
- Crimping tool
- Cable stripper/cutter
- Cable tester
- RJ45 connector boots/shields, if required

Difference Between Straight-Through and Crossover Cable

Straight-Through Cable

Both ends use the same wiring standard.

Example:

End A – T568A
End B – T568A

or

End A – T568B
End B – T568B

Crossover Cable

The two ends use different wiring standards.

End A – T568A
End B – T568B

T568A Colour Sequence

1. White-Green
2. Green
3. White-Orange
4. Blue
5. White-Blue
6. Orange
7. White-Brown
8. Brown

T568B Colour Sequence

1. White-Orange
2. Orange
3. White-Green
4. Blue
5. White-Blue
6. Green
7. White-Brown
8. Brown

Procedure

Step 1

The protective boot/shield was placed onto the Ethernet cable.

Step 2

Approximately 1.5 cm of the outer cable jacket was carefully removed using the stripping section of the crimping tool.

Step 3

The four twisted pairs were separated and untwisted. The eight individual wires were arranged according to the required T568A or T568B colour sequence.

Step 4

The wires were straightened and placed side by side in the correct order. The ends were trimmed evenly.

Step 5

The RJ45 connector was held with the contacts facing upward and all eight wires were inserted carefully while maintaining the correct order.

Step 6

The RJ45 connector was inserted into the crimping tool and pressed firmly to secure the wires to the connector contacts.

Step 7

The other end of the cable was prepared.

For a **straight-through cable**, the same wiring standard was used on both ends.

For a **crossover cable**, T568A was used on one end and T568B on the other end.

Step 8

The completed cable was tested using an Ethernet cable tester. The tester was checked to verify that all eight conductors were correctly connected.

Observation

The different types of network cables were studied and their characteristics, advantages, disadvantages and applications were understood. The Ethernet cable was prepared by arranging the wires according to the required colour sequence and attaching RJ45 connectors using a crimping tool.

The cable tester indicated that the conductors were properly connected.

Result

Thus, the different types of network cables—UTP, STP, coaxial and fibre optic cables—were studied. An Ethernet straight-through/crossover cable was successfully constructed using RJ45 connectors and a crimping tool, and its connectivity was tested successfully.